

DECATUR, IL, USA

WMO: 725316

Lat: 39.834N

Lon: 88.866W

Elev: 206

StdP: 98.88

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-14.0	-21.2	0.6	-15.7	-18.8	0.7	-13.7	12.4	-1.1	11.4	0.1	4.3	300	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.7	24.6	32.5	24.1	31.2	23.4	26.2	31.7	25.4	30.6	24.6	29.4	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.1	29.8	23.8	19.1	28.8	22.9	18.1	27.7	82.6	31.8	78.7	30.5	75.2	29.2	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.6	8.6	-20.7	35.8	3.9	1.5	-23.6	36.9	-25.8	37.8	-28.0	38.7	-30.9	39.8		
			-21.1	27.9	3.8	0.8	-23.8	28.5	-26.1	28.9	-28.2	29.4	-30.9	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-2.6	-0.2	5.5	11.9	17.7	22.6	24.3	23.3	19.8	13.1	6.0	0.2
	DBStd	10.71	6.84	6.41	6.14	5.13	4.68	3.50	3.13	2.99	4.13	5.06	5.54	6.04
	HDD10.0	1367	392	291	167	37	2	0	0	0	1	25	144	309
	HDD18.3	2986	648	519	398	201	71	7	1	2	31	176	369	563
	CDD10.0	2046	2	4	28	94	242	377	443	412	294	121	25	4
	CDD18.3	624	0	0	2	7	52	134	186	156	74	14	1	0
	CDH23.3	6095	0	0	12	79	463	1274	1905	1458	766	136	2	0
	CDH26.7	2217	0	0	1	11	126	466	765	538	278	32	0	0
Wind	WSAvg	4.3	5.1	5.1	5.1	5.3	4.4	3.7	3.1	3.0	3.2	4.1	4.8	4.7
Precipitation	PrecAvg	1003	57	51	67	100	118	118	105	86	78	81	88	54
	PrecMax	1303	154	123	139	216	259	235	322	136	209	181	208	110
	PrecMin	860	11	12	21	33	32	59	38	39	5	8	22	24
	PrecStd	124	33	28	31	47	65	51	66	26	53	43	43	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	18.6	25.1	28.0	31.9	34.0	36.2	35.0	33.9	30.2	22.7	17.6
		MCWB	14.3	13.3	17.7	18.7	22.4	23.9	25.9	25.1	22.4	21.1	16.0	15.3
	2%	DB	13.0	15.4	21.6	25.9	29.7	32.6	33.8	33.0	32.1	27.2	19.5	14.5
		MCWB	11.1	12.4	15.6	17.4	21.1	23.7	25.4	24.4	22.3	19.0	14.9	12.5
	5%	DB	10.2	12.3	18.5	23.3	27.9	31.3	32.4	31.5	29.9	24.4	17.4	11.7
		MCWB	8.2	9.5	13.7	16.1	20.2	23.3	25.1	23.9	21.3	17.5	13.8	9.7
	10%	DB	6.7	8.9	15.6	21.2	26.2	29.6	30.9	29.8	27.8	22.1	14.9	8.6
		MCWB	4.6	6.4	11.6	15.1	19.1	22.2	23.9	23.0	20.1	16.7	11.5	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.3	18.5	20.7	23.9	26.2	27.7	27.2	24.8	22.3	17.9	15.9
		MCDB	16.4	17.5	23.9	26.5	29.5	31.7	33.5	32.8	30.6	28.4	20.6	17.5
	2%	WB	11.7	12.6	16.6	18.9	22.5	25.2	26.6	25.8	23.7	20.6	16.1	12.8
		MCDB	12.8	14.8	20.8	23.2	27.9	30.9	32.1	31.1	29.4	25.0	18.7	14.4
	5%	WB	8.1	9.5	14.5	17.5	21.4	24.2	25.7	24.9	22.5	19.0	14.4	9.7
		MCDB	9.7	12.3	17.8	21.5	26.3	29.5	30.8	29.9	27.7	22.7	16.9	11.3
	10%	WB	4.7	6.6	11.9	15.9	20.2	23.3	24.9	24.0	21.4	17.3	12.0	6.9
		MCDB	6.6	8.9	15.1	20.2	24.5	28.1	29.7	28.4	26.0	21.3	14.6	8.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	10.4	11.7	11.1	10.8	10.9	11.3	12.9	11.6	9.5	7.7
		MCDBR	11.3	13.1	14.0	14.5	12.7	12.3	11.7	12.7	14.6	14.6	12.6	10.8
		MCWBR	9.7	10.2	9.2	8.4	6.3	5.7	5.4	5.9	6.5	7.9	9.1	9.1
	5% WB	MCDBR	10.7	12.5	12.5	12.3	11.3	11.0	10.6	10.8	12.0	12.0	11.4	10.0
		MCWBR	9.9	10.4	8.8	8.4	6.3	5.7	5.5	5.7	6.1	7.5	9.3	9.2
Clear-Sky Solar Irradiance	taub	0.303	0.315	0.352	0.392	0.436	0.435	0.460	0.453	0.411	0.360	0.312	0.297	
	taud	2.440	2.397	2.375	2.265	2.195	2.239	2.154	2.164	2.261	2.408	2.497	2.472	
	Ebn at Noon	865	904	902	882	847	845	820	815	827	840	849	845	
	Edh at Noon	80	96	110	131	144	138	149	144	122	95	76	72	
All-Sky Solar Radiation	RadAvg	1.83	2.69	3.66	4.80	5.47	6.28	6.20	5.51	4.66	3.22	2.15	1.54	
	RadStd	0.21	0.26	0.29	0.32	0.46	0.43	0.37	0.35	0.33	0.35	0.21	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.38	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+70

Nomenclature: See separate page